Lang Liu

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Education

Illinois Institute of Technology, Chicago, IL, United States
- Ph.D. in Computer Science, GPA: 3.89/4.00 — Jan 2024 - May 2027 (Expected)

Central South University, Changsha, Hunan, China
- B.S. in Civil Engineering, GPA: 86.9/100 — Sep 2013 - Jun 2017

Research Interests

Programming Languages, Formal Methods, Separation Logic, Program Verification, Compiler Security

Publications

Quantified Underapproximation via Labeled Bunches
L. Liu, F. Derakhshan, L. Jia, G. Moreno, M. Klein. OOPSLA 2025

End-to-End Guarantees for Compartmentalized Safety Critical Software
L. Liu, Z. Zhang, A. Vasudevan, F. Derakhshan, L. Jia, W. Mansky. Under Review

PIMSA: A Static Analysis Framework for Processing-in-Memory Programs
L. Liu, A. Madhu, S. Nawaz, F. Derakhshan, M. Zhou. Under Review

Work Experiences

Illinois Institute of Technology | Research Assistant & Teaching Assistant — Jan 2024 – Present
- Developed logical frameworks for compiler validation and the integration of formal verification and testing.
- Teaching Assistant for CS351 System Programming and CS534 Types and Programming Languages.

Amazon Web Services | Applied Scientist Intern — May 2026 – Aug 2026

Automated Reasoning Group
- Developed CranDSL, a Domain-Specific Language (DSL) in Lean 4 for writing test cases for the AWS IAM model.
- Leveraged Lean's dependent type system to enforce operation dependencies statically within test cases.
- Awarded Top 3 in AWS Identity Hackathon (Operational Excellence category) for building an LLM-aided test generation framework using CranDSL, achieving 100% line coverage for APIs considered.

Presence Inc | Founding Software Engineer — Apr 2022 –Aug 2023

A startup invested by BlueRun Ventures
- Developed backend services for several social apps from inception to launch, supporting over 10k DAU.
- Led the agile development process and CI/CD pipeline, ensuring weekly deployments to AWS/GCP and regular app releases on Google Play and Apple App Store, improving release efficiency by over 2x.
- Tech Stack: Java, Spring Boot, Vert.x, C#, Unity, Flutter, Redis, PostgreSQL, gRPC, AWS, GCP

Tencent Technology (Shenzhen) Co. Ltd | Senior Software Engineer — Jul 2021 – Apr 2022

Cloud & Smart Industries Business Group
- Led the "Reversed Instance" development, enabling more cost-effective cloud resource offerings.
- Optimized database queries, reducing production database CPU utilization to 1/20 of its original level.
- Tech Stack: Java, Spring Boot, Redis, MySQL/TDSQL, Kafka, Docker, Kubernetes, Tencent Cloud

WanGuoyou (Shenzhen) Technology Co. Ltd | Software Engineer ➜ Software Manager — Jul 2018 – Jul 2021
- Joined as a backend software engineer on the founding team to build an online travel agency (OTA) platform.
- Promoted to lead the technology department within 1.5 years, leading the team of 10+ engineers.
- Streamlined core workflows from flight booking to financial operations. Improved operational efficiency by 10x.
- Integrated APIs from 10+ suppliers (flights, trains, hotels, car rentals), significantly expanding product offerings.
- Tech Stack: Java, Scala, Spring Boot, Spring Cloud, Redis, MySQL, RocketMQ, Kubernetes, Alibaba Cloud

Honors & Awards

Oregon Programming Languages Summer School Fellowship | Jun 2024

National Olympiad in Informatics in Provinces (NOIP), Third Prize, Sichuan, China | Oct 2010